Using Credit Derivatives

Seminar Module

conducted by

ACF Consultants Ltd





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Why Use Credit Derivatives?

Credit Management and Investment

- Managing credit risk across a loan portfolio
  - Buying protection to reduce concentration risk
  - Selling protection to increase diversification
- Income generation and yield enhancement
  - e.g. Insurance company selling CDS for income and risk diversification
- Separating credit risk from the direct lending relationship
  - e.g. Bank using a CLO to securitise a loan portfolio

Trading

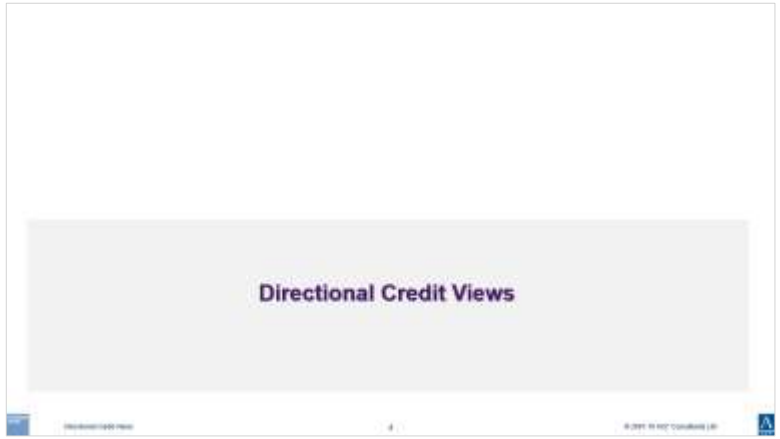
- Implementing directional credit views – negative and positive
- Monetising relative credit views
- Arbitrage opportunities
- Curve trades

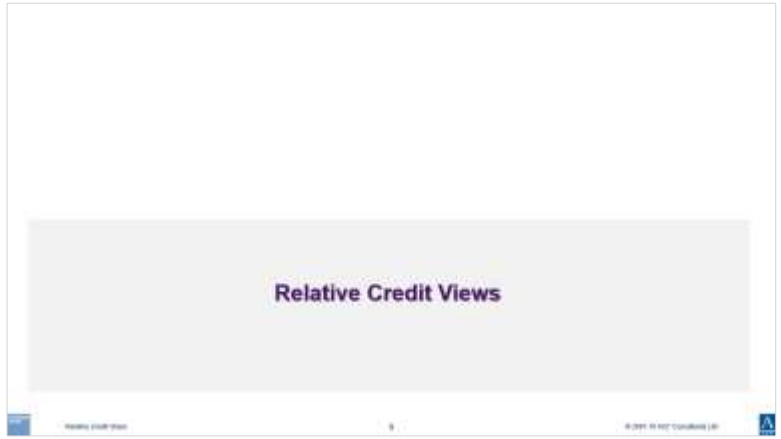
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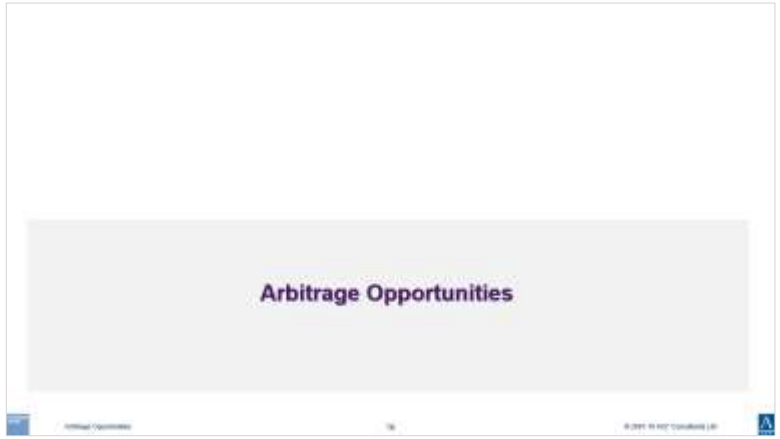
Key Point Summary

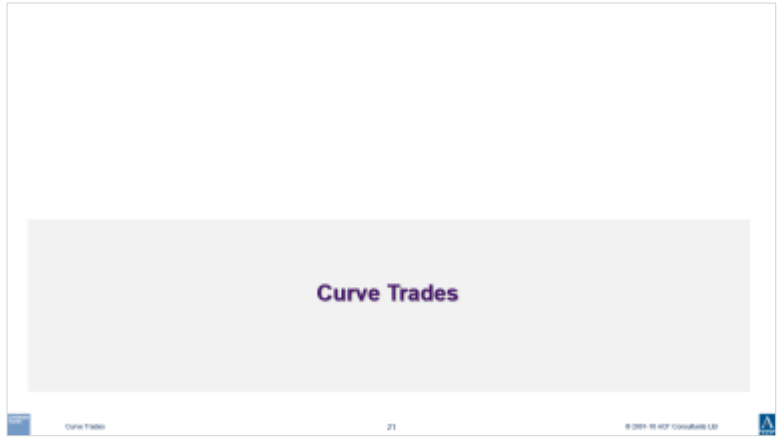
- CDS can be used for traditional purposes like
  - Reducing or eliminating credit risk (through buying protection)
  - Generating yield (through selling protection)
- CDS can also be used for trading purposes:
  - Directional credit views
  - Relative credit views
  - Arbitrage opportunities
  - Curve trades

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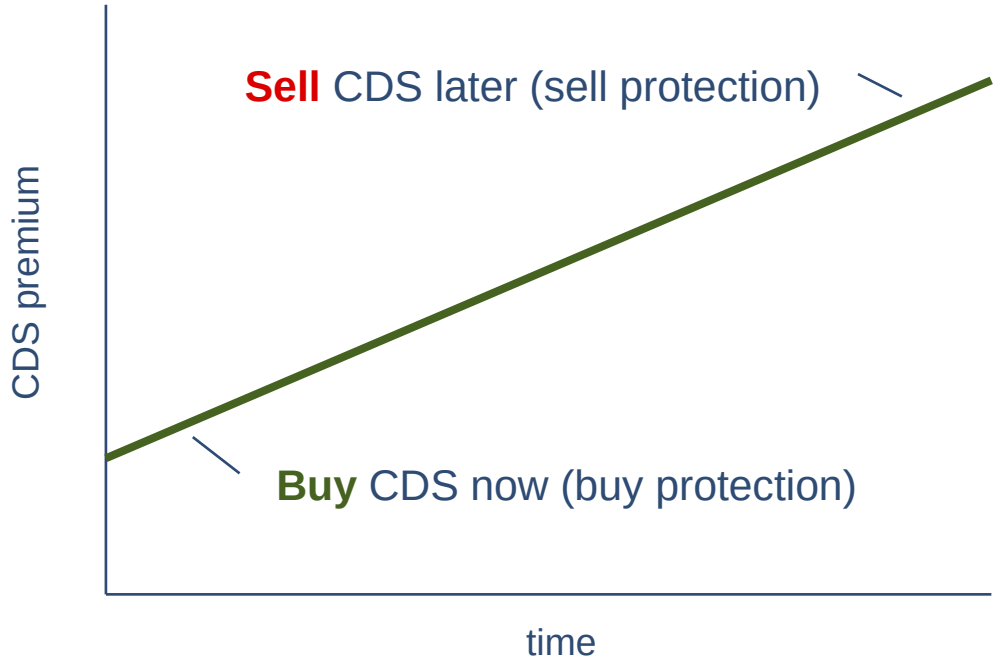
Directional Credit Views

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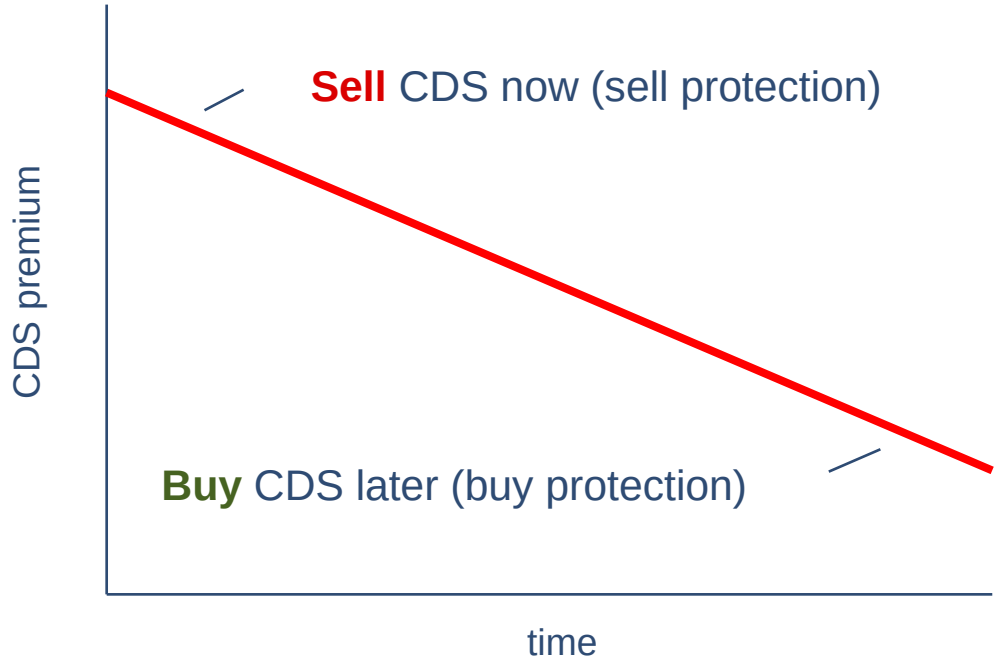
Directional Credit Views

- **Idea:** Credit spreads will increase (decrease) in absolute terms
- **Strategy:** Buy (sell) CDS to position accordingly

View: CDS premiums will increase



View: CDS premiums will decrease



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Directional Credit Views – Example

It is 26th December 2018.

- Following concerns around GE's financial prospects, 5yr CDS premiums on GE (in **blue**, on the left-hand axis) and its share price (in **red**, on the right-hand axis) have deteriorated. The situation was exacerbated when the major rating agencies downgraded GE from A to BBB+
- However, an investor believes that CDS spread will tighten in the months to come, as management is expected to defend GE's IG rating

Note the strong negative relationship between GE's equity price and its CDS premium.



Source: Goldman Sachs

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Directional Credit View – Results

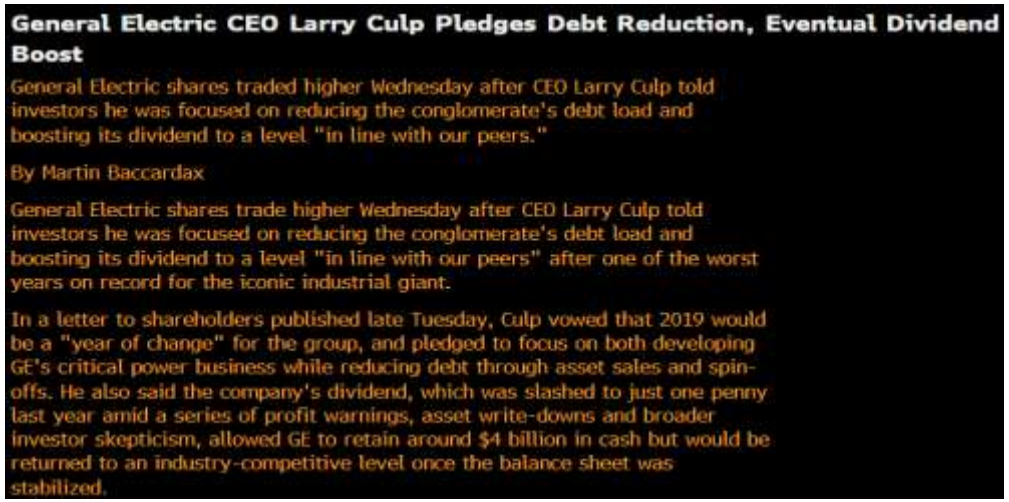
26 Dec 2018

- Investor believes that GE CDS spread will tighten
- Strategy: Sell protection on GE with $10m 5yr CDS (Dec 2023)
- In theory: Investor receives premium of 214 bp p.a.
- In practice:
  - Investor receives running premium of 100bp (GE is rated BBB+)
  - Investor receives $491,429 up-front, comprising:
    - $493,373 received (off-market adjustment)
    - $1,944 paid (7 days accrued)

4 Mar 2019

- GE premiums have indeed tightened to 89 bp!
- Strategy: Buy back protection on GE with $10m 5yr CDS (Dec 2023) to close out position
- In theory: Investor pays premium of 89 bp p.a

- In practice:
  - Investor pays running premium of 100bp
  - Investor receives $69,275 up-front, comprising:
    - $48,442 received (off-market adjustment)
    - $20,833 received (75 days accrued)



General Electric CEO Larry Culp Pledges Debt Reduction, Eventual Dividend Boost

General Electric shares traded higher Wednesday after CEO Larry Culp told investors he was focused on reducing the conglomerate's debt load and boosting its dividend to a level "in line with our peers."

By Martin Baccardax

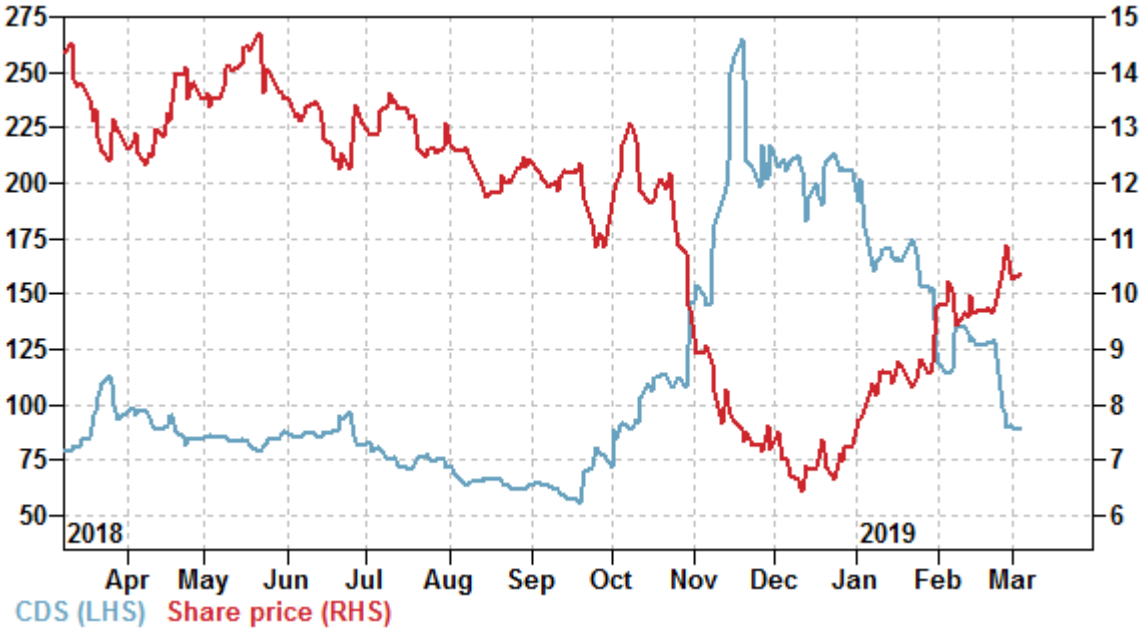
General Electric shares trade higher Wednesday after CEO Larry Culp told investors he was focused on reducing the conglomerate's debt load and boosting its dividend to a level "in line with our peers" after one of the worst years on record for the iconic industrial giant.

In a letter to shareholders published late Tuesday, Culp vowed that 2019 would be a "year of change" for the group, and pledged to focus on both developing GE's critical power business while reducing debt through asset sales and spin-offs. He also said the company's dividend, which was slashed to just one penny last year amid a series of profit warnings, asset write-downs and broader investor skepticism, allowed GE to retain around $4 billion in cash but would be returned to an industry-competitive level once the balance sheet was stabilized.

Analysis follows overleaf…

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Directional Credit View – Analysis



Profit Analysis

Actual profit:	$491,429 + $69,275 = $560,704

Source: Goldman Sachs

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Relative Credit Views

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Relative Credit Views

- **Idea:** The difference between two credit spreads will increase (decrease)
- **Strategy:** Buy one CDS and sell another CDS, to position accordingly

There are two types of spread structure:

Equal notionals

- The notional principal on both CDS are the same
- Structure is then carry-neutral (accrued premiums and actual coupons are identical)

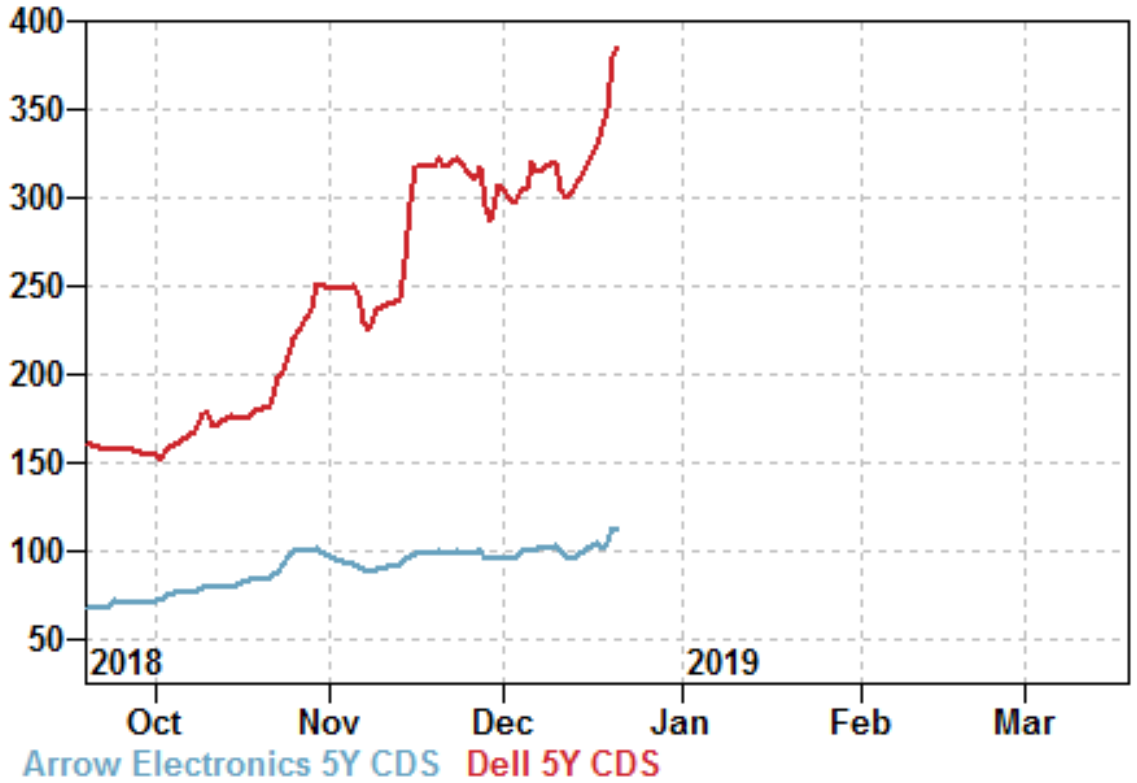
SDV01 neutral

- Notional principals are adjusted in the ratio of the SDV01s of the two CDS
- Structure is then immune to a parallel shift in credit spreads

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Relative Credit Views – Example

- Dell (**DELL**) and Arrow Electronics (**ARW**) both distributes and provides electronic products to their customers
- CDS on DELL have traded almost 275bp wider than those on ARW into the sell off late in Q4 2018
- An investor believes that DELL is being overly penalised in the CDS market, and believes that the spread difference between DELL and ARW will trade back to their 2y average range of 170bp…
- … but does not want exposure to the electronics sector as a whole

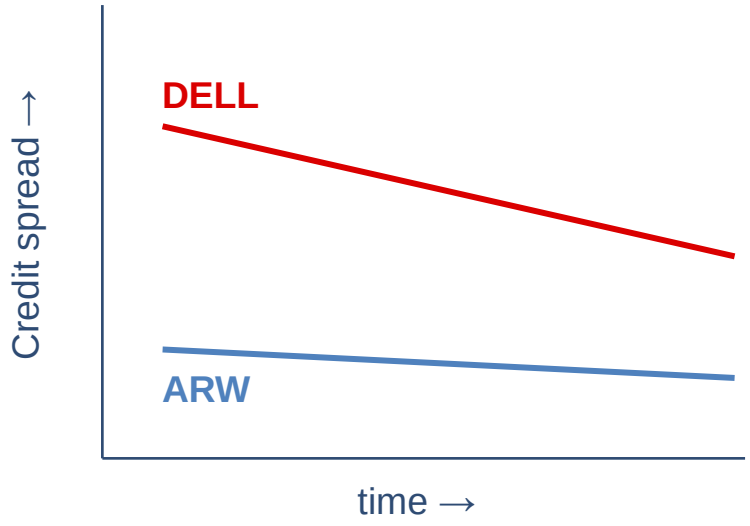


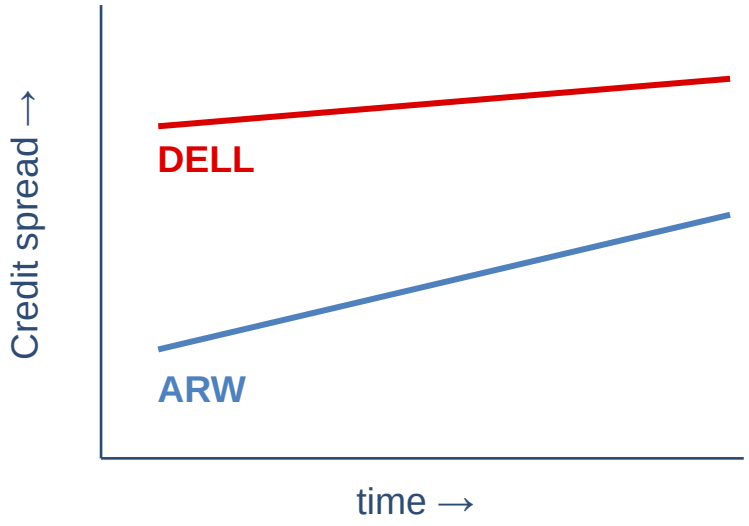
Source: Goldman Sachs

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Relative Credit Views – View

- **View:** CDS on the electronics sector could go either way over the next few months, but **DELL** will perform better than **ARW**





Scenario 1

- All electronics companies improve…
- …but **DELL** improves more than **ARW**

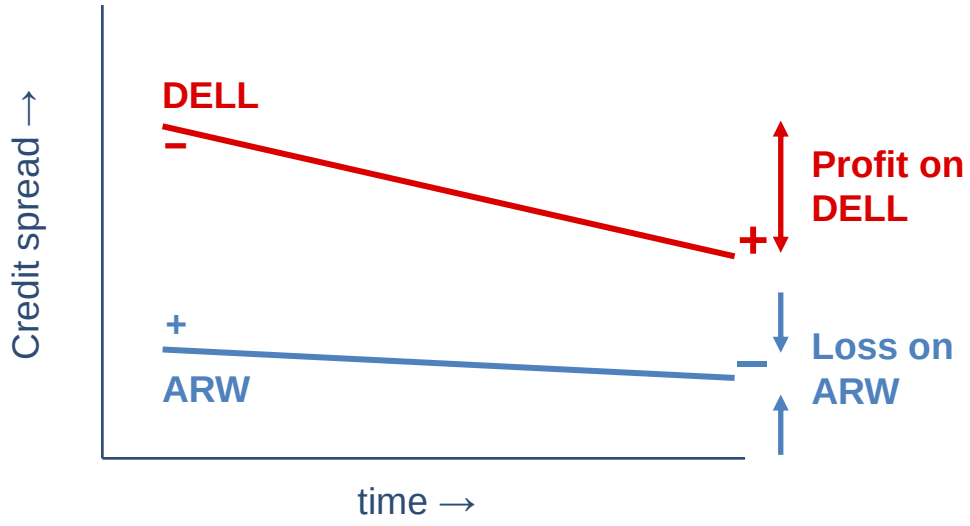
Scenario 2

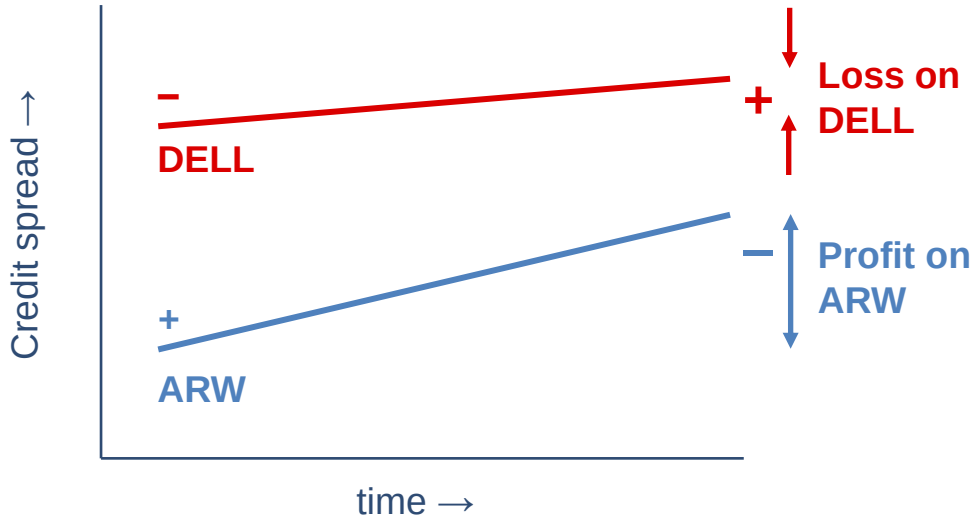
- All electronics companies worsen…
- …but **DELL** worsens less than **ARW**

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Relative Credit Views – Strategy

- **Strategy:** Sell **DELL** protection and Buy **ARW** protection





Scenario 1

- All electronics companies improve…
- …but **DELL** improves more than **ARW**

Scenario 2

- All electronics companies worsen…
- …but **DELL** worsens less than **ARW**

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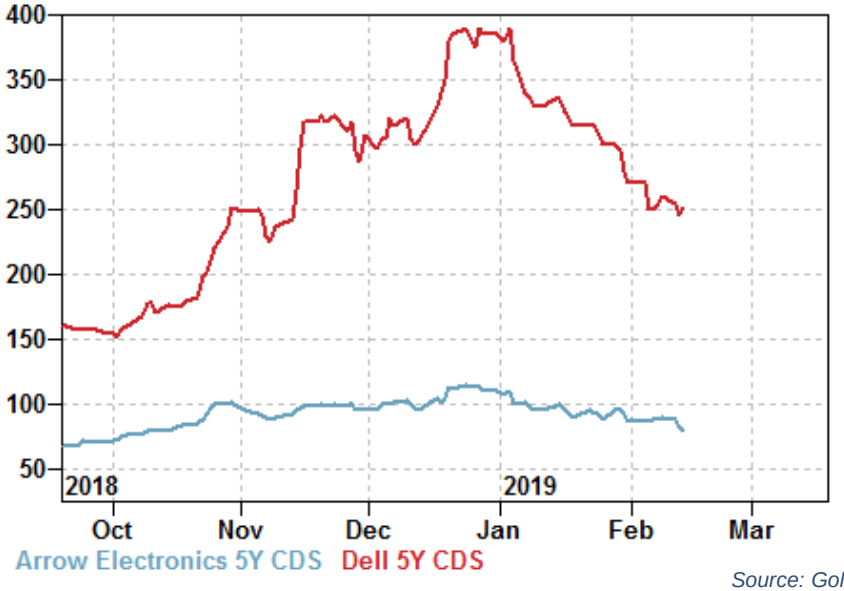
Relative Credit Views – What Happened…

On 24th Dec 2018, investor

- Sells $10m 5yr prot'n on **DELL** at 385bp (Dec 2023)
  - Investor receives running premium of 100bp
  - Investor receives $1,150,606 up-front as the off-market component
- Buys $10m 5yr prot'n on **ARW** at 112bp (Dec 2023)
  - Investor pays running premium of 100bp
    Investor pays $52,657 up-front as the off-market component:
- Investor receives $1,097,949 net upfront
- We can ignore the accrued coupon as these will cancel (equal notionals)

On 14th February 2019 the prices converges to their historical average, and investor:

- Buys $10m 5yr protection on **DELL** at 255bp
  - Investor pays $629,634 plus 57 days accrued
- Sells $10m 5yr protection on **ARW** at 81bp
  - Investor pays $84,685 plus 57 days accrued
- Investor pays $714,319 net upfront



Source: Goldman Sachs

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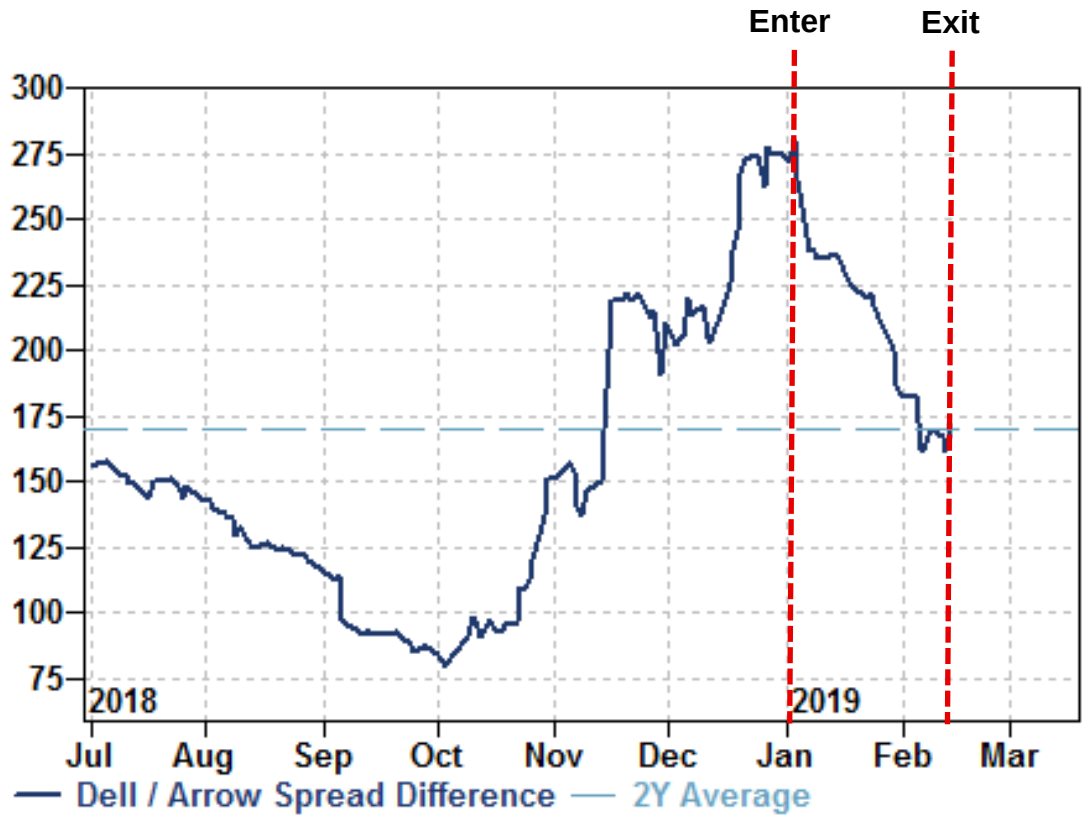
Relative Credit View – Summary

24th December 2018

- Investor believes that **DELL** CDS premiums will improve **relative** to **ARW**
- Sell $10m 5yr CDS on **DELL** at 385bp
- Buy $10m 5yr CDS on **ARW** at 112bp
- Investor **receives** $1,097,949 net upfront

14th February 2019

- DELL's CDS premiums have tightened by 130bp and the relationship has reverted to its historical average
- **DELL** CDS have fallen from 385bp to 255bp
- **ARW** CDS have also fallen from 112bp to 81bp
- Investor closes out positions and pays $730,152 net upfront



Profit Analysis	
Actual profit:	$1,097,949 – $714,319 = $383,630

Source: Goldman Sachs

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Arbitrage Opportunities

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Arbitrage Opportunities

- A number of financial products have embedded credit risk:
  - Bonds
  - Loans
  - Credit Derivatives
- Different markets can sometimes price credit risk differently
- If credit derivatives are available on these reference entities…
- … these inconsistencies can be exploited for a risk-free profit

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Arbitrage Trades

- **Idea:** Two credit instruments are priced inconsistently
- **Strategy:** Buy the cheaper instrument, and sell the dearer one

As before, there are two types of spread structure:

Equal notionals (the most common technique)

- The notional principal on both instruments are the same
- This is JTD-neutral, as the counterparty on both instruments is the same

SDV01 neutral

- Notional principals are adjusted in the ratio of the SDV01s of the two instruments
- Structure is then immune to a parallel shift in credit spreads

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Arbitrage Trades – Example

Weyerhaeuser (WY) **CDS** (in **red**) are trading rich to their bonds (in **blue**).

On 21st Sep 2018

- Weyerhaeuser 4.625% Sep 2023 is trading at 84bp on a **Z-spread** basis
- 5yr match-dated Weyerhaeuser **CDS** trading at 47bp
- Basis is therefore minus 37bp

Let's see if this difference can be exploited using a **negative basis trade**…



Weyerhaeuser

CDS Basis = Interpolated CDS Spread – Bond Z-spread

Source: Goldman Sachs

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Arbitrage Trades – Analysis

Strategy on 21st Sep 2018

- Borrow $10m at SOFR
- Buy $10m nominal of 5yr bond to yield S+84bp with par asset swap (Weyerhaeuser bond trading at 103.135)
- Buy $10m nominal of 5yr CDS protection on Weyerhaeuser at 47bp
- SDV01 is $485 for the CDS, and DV01 is $457 for the bond

Profit Analysis	
Carry	$10m × 37 bp = $37,000 p.a.

Weyerhaeuser 4.625% Sep 2023



Risk Analysis		
Annual net premium	$37,000 net income	
Spread risk	$2,800 loss ($485 × 10 × 10 – $457 × 10 × 10) $48,500 loss	(for a 10bp fall in both credit spreads) (for a 10bp fall in CDS relative to ASW)
JTD risk	$0m net result (Loss on bond: $60m; Gain on CDS: $60m)*	(if Weyerhaeuser defaults, 40% recovery)

* JTD risk will be exactly neutralized if maturity of CDS and Bond are matched – in this case, investor is takes on a net short jump risk

Source: Goldman Sachs

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Curve Trades

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Curve Trades

- **Idea:** The shape of the CDS curve will change (steepen / flatten)
- **Strategy:** Buy one part of the curve / sell another part of the curve

As before, there are two types of spread structure:

Equal notionals
(use this if default is considered a possibility)

- The notional principal on both CDS are the same.
- This is JTD-neutral

SDV01 neutral
(use this to immunise against parallel shift in credit curve)

- Notional principals are adjusted in the ratio of the SDV01s of the two CDS
- Structure is then immune to a parallel shift in the credit curve

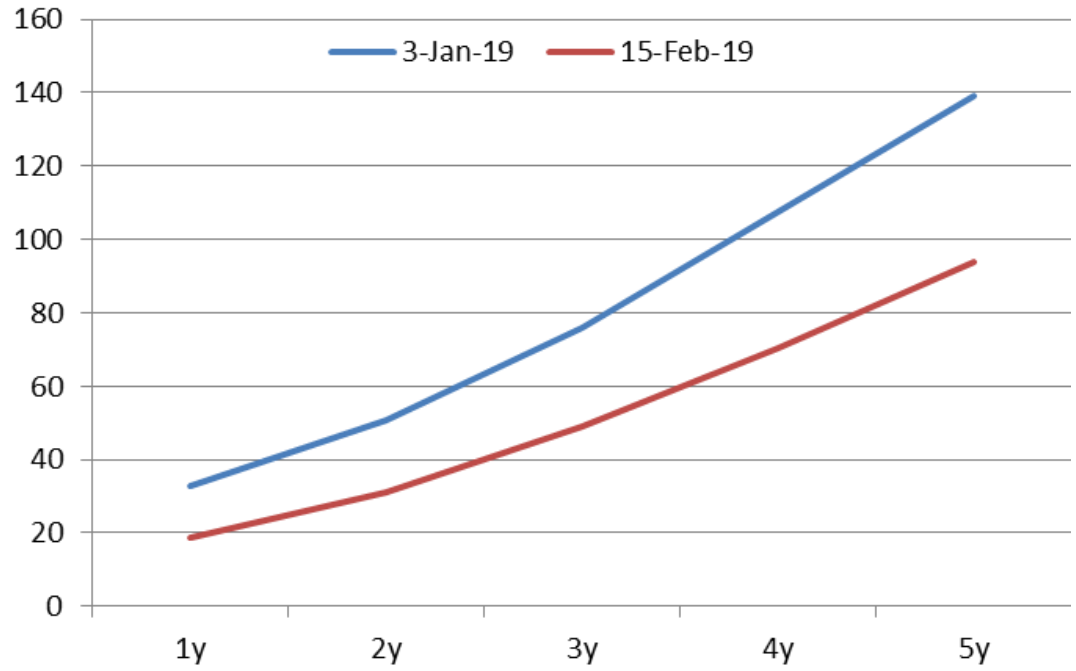
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Curve Trades – Example

- It is 3rd January 2019, and CDS premiums for AT&T have tightened, but the credit curve is still steep
- A hedge fund has no view on the level of AT&T credit, but believes that the credit curve will become flatter over the coming months in the 1yr-5yr range



CDS curves on AT&T



Source: Goldman Sachs

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Curve Trades – Analysis

Strategy on 3rd January 2019

- 1yr CDS: 33bp (SDV01 $96.1 / 1mm)
- 5yr CDS: 139bp (SDV01 $437.8 / 1mm)
- Buy 1yr protection at 33bp on $45m
  - Upfront received: $288,080 + $18,750
- Sell 5yr protection at 139bp on $10m
  - Upfront received: $173,545 – $4,167
- Net receipt upfront: $476,208

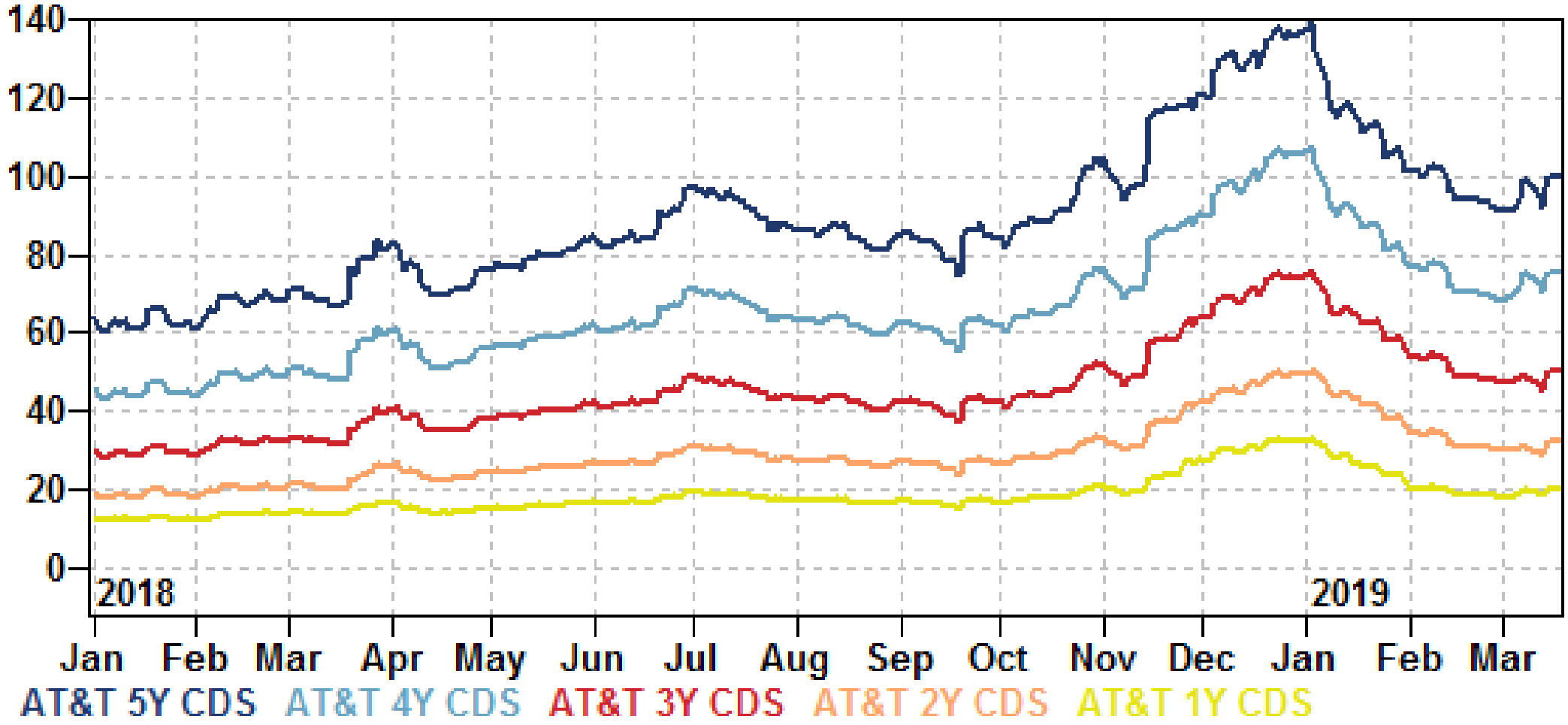
Strategy on 15th February 2019

- 1yr CDS: 19bp (SDV01 $84.6 / 1mm)
- 5yr CDS: 94bp (SDV01 $444.5 / 1mm)
- Sell back 1yr protection at 19bp on $45m
  - Upfront pay: $306,713 + $72,500
- Buy back 5yr protection at 94bp on $10m
  - Upfront receive: $26,615 + $16,111
- Net paid upfront: $336,487

Profit Analysis (10mm 1Y CDS vs 45mm 5Y CDS)		
Up-front payments:	$476,208 – $336,487 =	$139,721
Coupons paid:	$0 (trade dates are in between coupon dates)	
20 Dec 2019:	$0 (trade initiated after this date)	
20 Mar 2019:	$0 (trade exited before this date)	
Total		$139,721
Mark-to-market (1yr):	45 × (19 – 33) × ($96.1+$84.6)/2 =	–$56,920
Mark-to-market (5yr):	10 × (139 – 94) × ($437.8+$444.5)/2 =	+$198,517
Sub-total:	$198,517 – $56,920 =	$141,597
Carry:	[(10mm x 139bp) – (45mm x 33bp)] × 43/360 =	–$1,134
Total		$140,462 (approximated)

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Curve Trades – Analysis (cont'd)



Source: Goldman Sachs

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